INTERNSHIP REPORT

Data Analytics Job Simulation Internship

Organization	Interns of Industry
Internship	Data Analytics Job Simulation Internship
Student	Abhishek Kumar
Roll No.	2410030904
Semester / Section	5th Semester / Section 9
Session	2024–28
Duration	4 June 2026 – 2 July 2026

Submitted as part of the academic internship requirements

Internship Certificate

INTERNS OF INDUSTRY
THE FUTURE OF IOT LABS

CERTIFICATE
OF COMPLETION

This is to certify that

Abhishek Kumar

has successfully completed the Data Analytics Job Simulation Internship
at **Interns of Industry.**

Internship Duration: 4 June 2026 – 2 July 2026

During the internship, the candidate has completed practical tasks in:

- Exploratory data analysis and risk profiling
- Predicting delinquency with AI
- Business report and data storytelling for collections strategy
- Implementing an AI-driven collections strategy

Authorized Signatory
Interns of Industry

Enrollment Verification Code: IOI-DA-2026-ABHISHEK | User Verification Code: 7K2M9B4Q1Z | Issued by: Interns of Industry

The certificate is reproduced here as supporting documentation for the internship record.

Declaration

I, Abhishek Kumar (Roll No. 2410030904), declare that this internship report presents the work and learning associated with my Data Analytics Job Simulation Internship at Interns of Industry during 4 June 2026 to 2 July 2026. The report has been prepared for academic submission. Any illustrative code or examples included where original project artifacts are unavailable are identified as demonstration material.

Name: Abhishek Kumar

Roll No.: 2410030904

Acknowledgement

I would like to express my sincere gratitude to Interns of Industry for providing an opportunity to gain practical exposure to data analytics through a job simulation internship. I am thankful for the learning experience covering exploratory analysis, risk profiling, AI-based delinquency prediction, business reporting, data storytelling and an AI-driven collections strategy. I also thank my academic institution, faculty members and everyone who supported me during the internship and preparation of this report.

1. Introduction

Data analytics is the process of examining data to discover useful patterns, relationships and insights that can support decision-making. In a business environment, analytics can help organizations understand customer behaviour, identify risks, monitor performance and prioritize actions.

The Data Analytics Job Simulation Internship at Interns of Industry provided a project-oriented learning experience focused on applying analytical thinking to business problems. The internship covered exploratory data analysis and risk profiling, predicting delinquency with AI, business reporting and data storytelling, and implementing an AI-driven collections strategy.

2. Internship Objectives

- Understand an end-to-end data analytics workflow.
- Perform exploratory analysis and identify meaningful patterns in business data.
- Develop an understanding of risk profiling and prioritization.
- Explore AI / machine-learning approaches for delinquency prediction.
- Communicate analytical findings through business reports and data storytelling.
- Understand how analytics can support collections strategy and operational decisions.

3. Organization & Internship Overview

Organization: Interns of Industry

Role / Area: Data Analytics

Internship Type: Job Simulation Internship

Duration: 4 June 2026 – 2 July 2026

The internship was structured around practical business-oriented analytical tasks. Rather than focusing only on theory, the work emphasized understanding a problem, preparing and exploring data, interpreting risk, communicating insights and connecting analysis to possible business actions.

4. Task 1 – Exploratory Data Analysis & Risk Profiling

Exploratory Data Analysis (EDA) is used to understand the structure and characteristics of a dataset before deeper modelling. It may include checking data types, missing values, duplicate records, descriptive statistics, distributions and relationships between variables.

Risk profiling extends this analysis by identifying indicators associated with different levels of risk. In a collections context, the purpose is to help distinguish cases that may require greater attention from those that can be handled through routine processes.

Typical workflow:

- Load and inspect the dataset.
- Check data quality and missing values.

- Generate descriptive statistics.
- Visualize relevant distributions and relationships.
- Identify meaningful risk indicators.
- Summarize findings as business insights.

5. Task 2 – Predicting Delinquency with AI

Delinquency prediction is a classification-oriented analytics problem in which historical information can be used to estimate whether an account may fall into a delinquent category. A responsible workflow begins with data preparation and feature understanding before selecting and evaluating a suitable model.

A typical machine-learning workflow includes data cleaning, feature selection, train/test splitting, model training and evaluation. Depending on the dataset and requirements, classification algorithms such as logistic regression or tree-based methods can be explored.

Important: This report does not claim specific accuracy, precision, recall or other model metrics because those values should come from the actual internship dataset and model run. The code below is provided as a reusable demonstration template.

6. Demonstration Python Code

The following example shows a generic workflow for exploratory analysis and a binary classification model. Replace the filename and target column with the actual internship dataset and target field before use.

```
[illegible]
print(classification_report(y_test, pred))
```

7. Task 3 – Business Report & Data Storytelling

Data storytelling combines analysis, visualization and explanation to communicate what the data means for a business audience. A good business report should distinguish observations from assumptions and focus attention on insights that can influence decisions.

- Summarize key analytical findings.
- Use charts to make trends and comparisons easier to understand.
- Highlight important risk segments or patterns.
- Explain why a finding matters from a business perspective.
- Translate findings into practical recommendations.

8. Task 4 – AI-driven Collections Strategy

An AI-driven collections strategy uses analytics and predictive signals to prioritize collection efforts. The objective is not simply to predict risk, but to connect risk information with appropriate operational action.

- Identify accounts or cases using relevant risk indicators.
- Prioritize cases according to predicted or observed risk.
- Select suitable collection actions or communication approaches.
- Monitor outcomes and compare results over time.
- Use feedback and updated data to refine the strategy.

9. Tools & Technologies

Tool / Skill	Purpose
Python	Data analysis, automation and modelling workflows
Pandas	Data loading, cleaning and transformation
NumPy	Numerical operations
Matplotlib / visualization	Exploratory charts and data communication
Scikit-learn	Machine-learning workflow and evaluation
Data storytelling	Business-oriented presentation of analytical findings

10. Learning Outcomes

- Understanding of the end-to-end data analytics process.
- Improved ability to inspect, clean and interpret structured data.
- Practical understanding of exploratory analysis and risk profiling.
- Awareness of AI / machine learning for delinquency prediction.

- Improved data visualization and business storytelling.
- Better understanding of analytics-supported operational decision-making.

11. Conclusion

The Data Analytics Job Simulation Internship at Interns of Industry provided a practical introduction to applying analytics to business problems. The experience connected exploratory analysis and risk profiling with predictive thinking, business communication and collections strategy. It also reinforced the importance of data quality, appropriate modelling, clear visualization and responsible interpretation of results.

Going forward, continued practice with real datasets, SQL, statistics, Python, visualization and machine learning will help strengthen the skills developed during this internship.

12. References

- Internship certificate and internship task information provided by Interns of Industry.
- Python, Pandas, NumPy, Matplotlib and Scikit-learn documentation for the demonstration workflow.

Note: Specific project metrics, screenshots and dataset-derived findings should be added from the actual internship project artifacts if required by the institution.